

Statistical Analysis Results

Two-way ANOVA on TTC Perception Error — jamovi (v2.7)

ANOVA

Dependent variable: error_ttc

Source	Sum of Squares	df	Mean Square	F	p	η^2p
ttc_bin	712.92	10	71.29	40.15	< .001	0.457
weather_id	2.47	2	1.23	0.695	.499	0.003
ttc_bin × weather_id	24.03	20	1.20	0.677	.850	0.028
Residuals	846.85	477	1.78			

Post Hoc Tests

Pairwise comparisons — ttc_bin (Tukey correction)

Comparison	Mean Diff.	SE	df	t	p (Tukey)
1 – 2	-0.201	0.230	477	-0.873	.999
1 – 3	0.147	0.242	477	0.607	1.000
1 – 4	-0.023	0.223	477	-0.104	1.000
1 – 5	0.553	0.259	477	2.135	.553
1 – 6	1.364	0.330	477	4.134	.002
1 – 7	1.223	0.298	477	4.108	.002
1 – 8	1.650	0.482	477	3.425	.028
1 – 9	3.477	0.391	477	8.883	< .001
1 – 11	2.591	0.391	477	6.621	< .001
1 – 13	5.433	0.391	477	13.881	< .001
2 – 3	0.347	0.204	477	1.703	.834
2 – 4	0.177	0.181	477	0.979	.996
2 – 5	0.754	0.224	477	3.361	.034
2 – 6	1.564	0.303	477	5.159	< .001
2 – 7	1.423	0.268	477	5.315	< .001
2 – 8	1.850	0.464	477	3.989	.004
2 – 9	3.677	0.369	477	9.961	< .001
2 – 11	2.792	0.369	477	7.562	< .001
2 – 13	5.633	0.369	477	15.259	< .001
3 – 4	-0.170	0.196	477	-0.865	.999
3 – 5	0.407	0.237	477	1.717	.826
3 – 6	1.217	0.312	477	3.894	.005

3 – 7	1.076	0.278	477	3.866	.006
3 – 8	1.503	0.470	477	3.198	.056
3 – 9	3.330	0.377	477	8.836	< .001
3 – 11	2.444	0.377	477	6.486	< .001
3 – 13	5.286	0.377	477	14.026	< .001
4 – 5	0.577	0.217	477	2.651	.226
4 – 6	1.387	0.298	477	4.651	< .001
4 – 7	1.246	0.262	477	4.753	< .001
4 – 8	1.673	0.461	477	3.632	.014
4 – 9	3.500	0.365	477	9.586	< .001
4 – 11	2.614	0.365	477	7.161	< .001
4 – 13	5.456	0.365	477	14.945	< .001
5 – 6	0.810	0.326	477	2.484	.316
5 – 7	0.670	0.294	477	2.281	.449
5 – 8	1.097	0.479	477	2.288	.444
5 – 9	2.923	0.388	477	7.530	< .001
5 – 11	2.038	0.388	477	5.249	< .001
5 – 13	4.879	0.388	477	12.568	< .001
6 – 7	-0.141	0.357	477	-0.394	1.000
6 – 8	0.286	0.521	477	0.550	1.000
6 – 9	2.113	0.439	477	4.818	< .001
6 – 11	1.228	0.439	477	2.799	.162
6 – 13	4.069	0.439	477	9.279	< .001
7 – 8	0.427	0.501	477	0.852	.999
7 – 9	2.254	0.415	477	5.432	< .001
7 – 11	1.368	0.415	477	3.298	.041
7 – 13	4.210	0.415	477	10.146	< .001
8 – 9	1.827	0.562	477	3.252	.047
8 – 11	0.941	0.562	477	1.675	.848
8 – 13	3.783	0.562	477	6.734	< .001
9 – 11	-0.886	0.487	477	-1.820	.768
9 – 13	1.956	0.487	477	4.020	.003
11 – 13	2.842	0.487	477	5.841	< .001

Note. Comparisons based on estimated marginal means.

References

- [1] The jamovi project (2025). jamovi (Version 2.7) [Computer Software]. <https://www.jamovi.org>
- [2] R Core Team (2025). R: A Language and environment for statistical computing (Version 4.5) [Computer software]. <https://cran.r-project.org>

- [3] Fox, J., & Weisberg, S. (2024). car: Companion to Applied Regression [R package]. <https://cran.r-project.org/package=car>
- [4] Lenth, R. (2025). emmeans: Estimated Marginal Means, aka Least-Squares Means [R package]. <https://cran.r-project.org/package=emmeans>